| MODULE NAME: | MODULE CODE: |
| --- | --- |
| CLOUD DEVELOPMENT B | CLDV7112/w |

**ASSESSMENT TYPE: PROJECT 2 (PAPER AND MARKING RUBRIC)**

**TOTAL MARK ALLOCATION: 100 MARKS**

**TOTAL HOURS: A MINIMUM OF 15 HOURS IS SUGGESTED TO COMPLETE THIS ASSESSMENT**

*By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.*

**INSTRUCTIONS:**

1. ***No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
2. ***Make a copy of your assignment before handing it in.***
3. *Assignments must be typed unless otherwise specified.*
4. *Begin each section on a new page.*
5. *Follow all instructions on the cover sheet.*
6. *This is an individual assignment.*
7. *Answer All Questions.*
8. ***Instructions for submitting your assessment:***
   a. *Use of good programming practice and comments in code is compulsory.*
   b. *Save your solution/project files in your GitHub repository for this module.*
   c. *Save all files (including any source code files, template files, design files, image files, text files, database files, etc.) within your GitHub repository.*
   d. *Do NOT save zipped (archive) files in your GitHub repository unless specifically instructed to do so.*
   e. ***Important:*** *Upon completion of your assessment, you must save and close all your open files before submitting your work. You will submit your assessment on the LMS page for this module.*
   f. ***To complete your submission:*** *Create a document in MS-Word or Notepad. The document name must follow the format explained in the instructions.*
   g. *In this document, include the following: Your student number, the module code, all required answers,* ***the link to your GitHub repository*** *where you saved your practical work, and the URL of your web application.*
   h. *Submit this document in the LMS, using the submission links for this module.*

## Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate.** The examples provided are not exhaustive but illustrate the errors.

**Deductions**

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage.**
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage.**
- **Where both minor and major errors** (e.g., two minor and three majors, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

| **Required:**<br>**Consistent and congruent referencing** | **Minor errors**<br>**Deduct 5% from the overall percentage.**<br>**Example: if the response receives 70%, deduct 5%. The final mark is 65%.** | **Major errors**<br>**Deduct 10% from the overall percentage.**<br>**Example: if the response receives 70%, deduct 10%. The final mark is 60%.** |
|---|---|---|
| **Consistency**<br>• The **correct referencing style** for the **discipline** – i.e., either **Harvard**, OR **APA** (for Psychology)**,** OR **Law**, OR **IEEE** (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list.<br>• **Concepts** and **ideas** that are **quoted** and/or **paraphrased** are referenced consistently throughout.<br>• **Position of the in-text reference**: an in-text reference is positioned consistently where appropriate for every quote and paraphrase. | **Minor inconsistencies:**<br>• The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list.<br>• For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author's name in one bibliography/reference list entry, and after the source title in another, etc.<br>• Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work.<br>• Position of the references: in-text references are only given at the beginning and/or end of every paragraph. | **Major inconsistencies:**<br>• Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list.<br>• Multiple referencing styles for the same source types have been used.<br>• For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances.<br>• Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text.<br>• Position of the references: in-text references are only given at the beginning or end of large sections of work. |
| **Feedback on referencing consistency:** | | |
| **Congruency**<br>• **Each source** reflected within **in-text references** is included accurately in the **bibliography/reference list.**<br>• All **bibliography/reference list** entries are in the **required order** for the referencing style used (e.g., alphabetical, alphabetical under subheadings, numerical).<br>• All **direct quotes** and **paraphrases** have been **integrated appropriately** into the text using introductory phrases, accurate grammar, etc. | **Minor incongruences:**<br>• There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g., publication year, city of publication, URL date accessed, etc.).<br>• There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order.<br>• An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could. | **Major incongruences:**<br>• No relationship/several incongruencies between the in-text referencing and the bibliography/reference list.<br>• For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc.<br>• Most sources are listed in a haphazard order throughout the bibliography/reference list.<br>• Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text. |
| **Feedback on referencing congruency:** | | |
| **Overall feedback on referencing, with suggested improvements:** | | |

 

**Background**

ABC Retail, a rapidly growing online retailer, currently manages its order processing system using an aging **on-premises** infrastructure. They store customer orders and product information in a **traditional relational database** system, which struggles to handle the increasing volume of transactions during peak shopping seasons like Christmas and other related holidays.

Additionally, the company stores product images in **network shared drives**, leading to storage inefficiencies and slow access times. The message queuing system, powered by **legacy middleware**, lacks scalability and reliability, often resulting in message delivery delays and processing errors. As a result, ABC Retail is facing customer complaints, missed sales opportunities, and operational inefficiencies.

Despite migrating their order processing system to the cloud, ABC Retail continues to face challenges in handling **real-time event processing** and reliable **message queuing**. Their current setup relies on a combination of custom-built event processing pipelines and third-party messaging solutions, which lack the scalability and flexibility needed to support the company's growing business demands. As a result, ABC Retail experiences delays in order processing, inconsistent messaging delivery, and difficulty in maintaining and scaling their existing infrastructure.

ABC Retail's current **data analytics** infrastructure struggles to keep pace with the growing volume and complexity of customer data. **Traditional relational databases** and **on-premises analytics tools** are unable to efficiently process and analyse diverse data types, leading to delays in generating actionable insights. As a result, ABC Retail faces challenges in personalizing customer experiences, optimizing product recommendations, and improving operational efficiency.

**<u>To work on this Project, students are required to</u>**:

- Have access to an Azure account with available credit. This access will be arranged by your lecturer/campus at the start of this module.
- Use Microsoft Visual Studio for your coding.
- Save source code in a GitHub repository. This access will be arranged by your lecturer/campus at the start of this module.

**<u>The submission of this Project will require you to do the following</u>**:

- Create a **document** in MS Word which contains the following:
  - Your student number.
  - The module code.
  - All answers required, including typed answers, diagrams, and screenshots.
  - The ***URL of the Web App*** that you developed.
- The document name must follow the format shown here:
  - ***StudentNumber_ModuleCode_Project#.***
  - E.g., if your student number is *12345* and you are submitting *Project 1* for the module *PROG121*, create a document named ***12345_Prog121_Project1.***
- Submit this document in the LMS, using the submission link on the LMS page for this module.

**Summary Sheet**

| Item | Description |
| --- | --- |
| **Summary of Activities** | The student needs to submit this Project on the Learn/LMS page for this module.<br><br>***NB. Please follow the given instructions to supply a document that includes the needed answers, screenshots, the URL of the Web App module developed, and the link to the GitHub repository containing your project source code.*** |
| **Tools and Resources** | • Microsoft Visual Studio.<br>• Microsoft SQL Server.<br>• Microsoft Word or other word processing software.<br>• Windows Azure Portal.<br>• Windows Azure subscription with Microsoft Azure Storage.<br>• A GitHub repository. |
| **Calculation of Marks** | The final module mark will be weighted as follows:<br>• Project 1 – 25% (formative)<br>• Project 2 – 30% (formative)<br>• ICE Tasks – 10%<br>• Practicum – 35% (summative) |

**Project 2 — Integrating Azure Services into a Web Application (Marks: 100)**

Related Content: Learning Units: 4 – 6

**Assessment**

| Assessment/Deliverable | Marks | Weight | Duration |
|---|---|---|---|
| Project 2 | 100 | 30% | 15 hours |

**Your submission document**

Your submission for this Project must be submitted in an MS-Word document, containing the following:

- Your student number.
- The module code.
- The URL of your deployed application (using the format http://student_number.azurewebsites.net)
- The GitHub link for the web application source code.
- Screenshots showing your successful implementation of the functionalities as described below, as well as screenshots of the deployed web application
- Written answers for discussion questions.

Note: *Make sure you read the instructions given earlier in this document and follow the specified file name format.*

**A. Integrating Functions to build a robust application architecture**

You are required to expand on the application you developed for Project 1. You must integrate four functions into your code to enhance the scalability, cost-effectiveness, and cloud suitability. The four functions will call the services used in your existing application:

- Store information in Azure tables.
- Write to Blob Storage.
- A queue is written to/from for transaction information.
- Write to Azure files.

**B. Using services for improving the customer experience**

Discuss how the following services could add value to the customer experience in your app:

- Azure Event Hubs.
- Azure Event Bus.

 

Once you have tested your web application, you are required to deploy it to an Azure App Service and make it accessible through a link. Ensure that this has been tested both on your local computer and on the App Service.

Once you have completed your submission document for this Project, submit your document using the submission link on the Learn page for this module.

Note: *Make sure you read the instructions given earlier in this document and follow the specified file name format.*

| **Marking Rubric (Project 2)** | | | | | **Comments** |
|---|---|---|---|---|---|
| | **Excellent** | **Good** | **Developing** | **Poor** | |
| Create a function that stores information in Azure tables<br>• Screenshots of the function on the Azure function app and code shared. | 16 – 20 | 11 – 15 | 6 – 10 | 0 - 5 | |
| Create a function that writes to Azure blob storage<br>• Screenshots of the function on the Azure function app and code shared. | 16 – 20 | 11 – 15 | 6 – 10 | 0 - 5 | |
| Create a function that reads from/writes to the Azure queue<br>• Screenshots of the function on the Azure function app and code shared.<br>• Screenshot of the message in the queue | 16 – 20 | 11 - 15 | 6 – 10 | 0 - 5 | |
| Create a function that sends a file to Azure Files<br>• Screenshots of the function on the Azure function app and code shared.<br>• Screenshot of the file in Azure files | 16 – 20 | 11 – 15 | 6 – 10 | 0 - 5 | |
| • Discussed the 2 services under the following headings:<br>o Description of service<br>o Mechanism<br>o How it adds value to end users | 16 – 20 | 11 - 15 | 6 – 10 | 0 - 5 | |
| **Project 2 Total** | | | | | **/100** |

**[TOTAL MARKS: 100]**